

Refutation of conjecture 00000000416

The orbit count of evacuation on tableaux of shape (n, n) is inconsistent at $n = 2$

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Abstract

Conjecture 00000000416 asserts that, for the evacuation operator acting on the standard Young tableaux of two-row shape (n, n) , the number of orbits is 2^{n-1} , every orbit length divides $2n$, and the number of orbits of length exactly 2 is the Fibonacci number F_{n-1} . We show that the first and third assertions are jointly impossible already at $n = 2$: they require at least three tableaux, whereas the shape $(2, 2)$ carries exactly two. The argument uses nothing about evacuation beyond the fact that an orbit is nonempty; in particular it does not use that evacuation is an involution.

1 The conjecture

The statement under consideration, verbatim from the corpus, is the following.

Definition: The evacuation operator e acts on standard Young tableaux of two-row shape (n, n) ; its orbits are the cycles of the iterates of e . Conjecture: The number of orbits is 2^{n-1} , every orbit length divides $2n$, and the number of orbits of length exactly 2 is the Fibonacci number F_{n-1} .

Write T_n for the set of standard Young tableaux of shape (n, n) , and let

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

denote the n -th Catalan number. It is classical that $|T_n| = C_n$: a standard tableau of shape (n, n) is determined by the set of entries placed in the top row, and the ballot condition says exactly that this set encodes a Dyck path of semilength n .

The conjecture makes three simultaneous claims about the action of e on T_n :

$$\#\{\text{orbits of } e\} = 2^{n-1}, \tag{1}$$

$$\text{every orbit length} \mid 2n, \tag{2}$$

$$\#\{\text{orbits of length exactly } 2\} = F_{n-1}. \tag{3}$$

2 The contradiction at $n = 2$

Lemma 1. *The shape $(2, 2)$ carries exactly two standard Young tableaux.*

Proof. The entries are 1, 2, 3, 4. The top-left cell is smaller than both its row neighbour and its column neighbour, and the bottom-right cell is larger than both its row neighbour and its column neighbour; since the top-left entry is below the bottom-right entry along the top row, the top-left entry is smaller than all three others, hence equals 1, and symmetrically the bottom-right entry equals 4. The remaining entries 2 and 3 fill the two off-diagonal cells, and each of the two placements is admissible. Explicitly,

$$\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \quad \text{and} \quad \begin{array}{cc} 1 & 3 \\ 2 & 4 \end{array}$$

are the only possibilities. (Equivalently $|T_2| = C_2 = 2$.) □

Theorem 1. *Conjecture 00000000416 is false.*

Proof. Suppose the conjecture holds at $n = 2$. By (1) there are $2^{2-1} = 2$ orbits in total. By (3), the number of orbits of length exactly 2 is $F_{2-1} = F_1 = 1$.

The orbits partition T_2 . One orbit therefore contains exactly 2 tableaux, and the second orbit is nonempty, so it contains at least 1. Hence

$$|T_2| \geq 2 + 1 = 3.$$

But $|T_2| = 2$ by Lemma 1. This is a contradiction.

Therefore the conjunction of (1) and (3) fails at $n = 2$, and with it the conjecture. □

3 What the argument does and does not use

The proof of Theorem 1 uses exactly one property of evacuation: that its orbits are nonempty. It does *not* use that e is an involution, nor that orbit lengths are 1 or 2, nor any information about how e acts on a particular tableau. This is deliberate — it makes the refutation robust against any disagreement about the precise definition of evacuation, since any map whatsoever whose iterates partition T_2 into two orbits, one of size 2, would require $|T_2| \geq 3$.

Remark 1 (A second, independent failure). If one additionally uses that evacuation is an involution, so that every orbit has length 1 or 2, then (1) and (3) force

$$|T_n| = (2^{n-1} - F_{n-1}) \cdot 1 + F_{n-1} \cdot 2 = 2^{n-1} + F_{n-1}.$$

For $n = 4$ the left side is $C_4 = 14$ while the right side is $8 + 2 = 10$, so the conjecture fails again, independently of the $n = 2$ case. (For $n = 3$ the identity $5 = 4 + 1$ does hold, and for $n = 1$ the identity $1 = 1 + 0$ holds as well; the failures begin at $n = 2$ and resume from $n = 4$.)

Remark 2 (On claim (2)). We make no assertion about (2). It is consistent with the others: orbit lengths are at most 2 when e is an involution, and both 1 and 2 divide $2n$ for $n \geq 1$. The refutation rests entirely on the joint inconsistency of (1) and (3).

4 Formalisation

The accompanying Lean 4 project formalises the argument in `lean4/Main.lean`. It contains:

- an exhaustive enumeration of the $4^4 = 256$ fillings of the diagram $(2, 2)$ by the letters $0, 1, 2, 3$, filtered to the standard ones. The theorem `sols_length` states that exactly two survive, and is proved by `rfl`: the kernel performs the search and checks the count, so the theorem depends on *no* axioms at all;
- the counting lemma `orbit_bound`: if a list is split into two nonempty parts and one part has at least two elements, the total is at least 3;
- `refutation_at_two`: no such splitting of the tableaux of shape $(2, 2)$ exists.

The project uses only the Lean 4 core (no Mathlib) and contains no `sorry`. The axioms used are `propext` and `Quot.sound`, inherited from the standard treatment of lists.

5 Reproducing the enumeration

The accompanying `reproduce.py` (standard library only) enumerates the standard Young tableaux of shape (n, n) for small n , reports C_n alongside the quantity $2^{n-1} + F_{n-1}$ required by the conjecture, and prints the tableaux of shape $(2, 2)$. Running it reproduces the numbers used above.